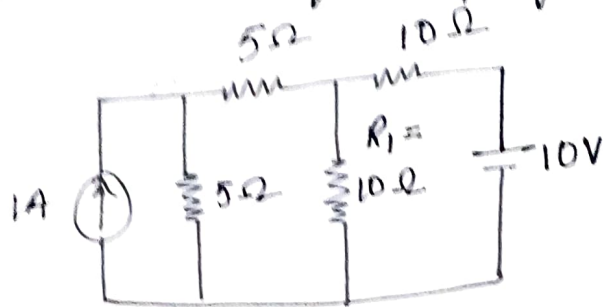
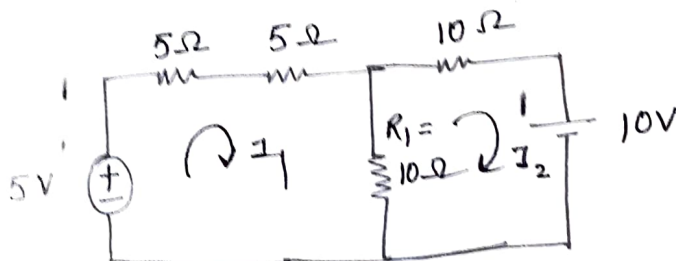


Q4. Using mesh eq<sup>n</sup> method find the current in resistance  $R_1$  of the n/w given below.



Sol<sup>n</sup> 1A current source in ||<sup>d</sup> with  $5\Omega$   
Apply source conversion.



KVL in mesh (1)

$$-5 + 10I_1 + 10(I_1 - I_2) = 0$$

$$-5 + 20I_1 - 10I_2 = 0$$

$$20I_1 - 10I_2 = 5$$

$$4I_1 - 2I_2 = 1 \quad \text{--- (1)}$$

KVL in mesh (2)

$$10I_2 + 10 + 10(I_2 - I_1) = 0$$

$$10I_2 + 10 + 10I_2 - 10I_1 = 0$$

$$20I_2 - 10I_1 = -10$$

$$2I_2 - I_1 = -1$$

$$2I_2 + 1 = I_1 \quad \text{--- (2)}$$

Put  $I_1$  in eq (1)

$$4(2I_2 + 1) - 2I_2 = 1$$

$$8I_2 + 4 - 2I_2 = 1$$

$$6I_2 + 4 = 1$$

$$6I_2 = -3$$

$$I_2 = \frac{-3}{6} = -\frac{1}{2}$$

Put  $I_2$  in eq (2)

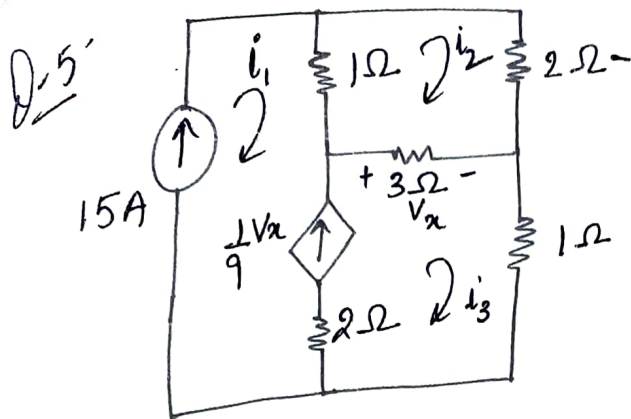
$$2\left(-\frac{1}{2}\right) + 1 = I_1$$

$$I_1 = -1 + 1 = 0$$

Current through  $10\Omega$

$$I_1 - I_2 = 0 - \left(-\frac{1}{2}\right) = \frac{1}{2} = 0.5A$$

$$\therefore I_2 - I_1 = -\frac{1}{2} - 0 = -\frac{1}{2} = -0.5A$$



Soln

$$i_1 = 15A \text{ --- (1)}$$

⇒ Apply KVL in mesh (2)

$$2i_2 + 3(i_2 - i_3) + 1(i_2 - i_1) = 0$$

$$2i_2 + 3i_2 - 3i_3 + i_2 - i_1 = 0$$

$$6i_2 - 3i_3 - i_1 = 0 \text{ --- (1)}$$

$$\Rightarrow -V_x + 3(i_3 - i_2) = 0$$

$$V_x = 3(i_3 - i_2) \text{ --- (2)}$$

$$\Rightarrow \frac{1}{9}V_x = i_3 - i_1 \text{ --- (3)}$$

$$\Rightarrow i_1 = 15A \text{ --- (4)}$$

Put  $V_x$  from eqn (3) in eqn (2).

$$9(i_3 - i_1) = 3(i_3 - i_2)$$

$$9i_3 - 9i_1 = 3i_3 - 3i_2$$

$$9i_3 - 3i_3 = -3i_2 + 9i_1$$

$$6i_3 = -3i_2 + 9i_1$$

$$2i_3 = -i_2 + 3i_1 \text{ --- (5)}$$

Put  $i_1$  from eqn (1) in eqn (5)

$$2i_3 = -i_2 + 3(15)$$

$$2i_3 = 45 - i_2 \text{ --- (6)}$$

$$i_2 = 45 - 2i_3$$

Put  $i_2$  from eqn (6) in eqn (1)

$$6(45 - 2i_3) - 3i_3 - i_1 = 0$$

$$270 - 12i_3 - 3i_3 - 15 = 0$$

$$255 - 15i_3 = 0$$

$$i_3 = \frac{255}{15} = 17A$$

Put  $i_3$  in eqn (1)

$$6i_2 - 3(17) - 15 = 0$$

$$6i_2 - 51 - 15 = 0$$

$$6i_2 - 66 = 0$$

$$i_2 = \frac{66}{6} = 11A$$

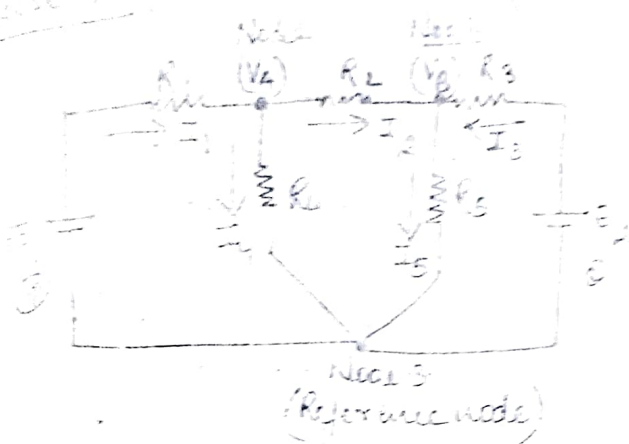
$$i_2 = 11A$$

(E)

Example 1

Find voltage

Given:



Find voltage across  $R_3$  and  $R_2$  in series.

Consider the circuit has three nodes. One of these nodes is considered as reference node or datum node.

Let node 3, KCL at node 1,

$$I_1 = I_2 + I_3$$

$$\frac{E_1 - V_1}{R_1} = \frac{V_1 - V_2}{R_2} + \frac{V_1 - V_3}{R_3}$$

$$\frac{E_1}{R_1} - \frac{V_1}{R_1} = \frac{V_1}{R_2} - \frac{V_2}{R_2} + \frac{V_1}{R_3} - \frac{V_3}{R_3}$$

$$\frac{E_1}{R_1} - \frac{V_1}{R_1} - \frac{V_1}{R_2} + \frac{V_2}{R_2} - \frac{V_1}{R_3} + \frac{V_3}{R_3} = 0$$

For node 2,

$$I_2 = I_3$$

$$\frac{V_1 - V_2}{R_2} = \frac{V_2 - V_3}{R_3}$$

$$\frac{V_1 - V_2}{R_2} + \frac{V_2 - V_3}{R_3} = \frac{V_2 - V_3}{R_3}$$

$$\frac{V_1}{R_2} - \frac{V_2}{R_2} + \frac{V_2}{R_3} - \frac{V_3}{R_3} = \frac{V_2 - V_3}{R_3}$$

$$\frac{V_1}{R_2} + \frac{V_2}{R_3} - \frac{V_2}{R_2} - \frac{V_3}{R_3} = 0$$

$$V_1 \left[ \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_3} \right] - \frac{V_2}{R_2} - \frac{V_3}{R_3} = 0 \quad \text{--- (1)}$$

from

Eqn (1)

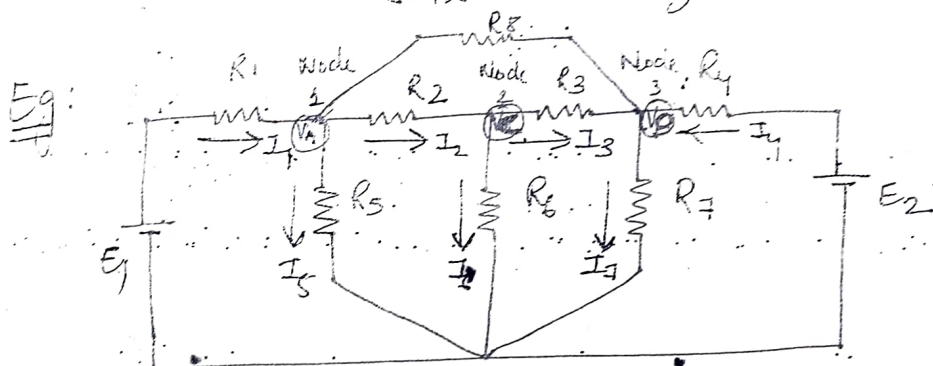
- 1) The product of node potential  $V_A$  and  $\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4}\right)$  i.e. the sum of the reciprocals of the branch resistance connected to this node.
- 2) Minus the ratio of adjacent potential  $V_B$  and the interconnecting resistance  $R_2$ .
- 3) Minus ratio of adjacent battery voltage  $E_1$  and interconnecting resistance  $R_1$ .
- 4) All the above set to zero.

Sol. for eqn (2)

Eqn (1) & (2) can be written as,

$$V_A(G_1 + G_2 + G_4) - V_B \cdot G_2 - E_1 G_1 = 0$$

$$V_B(G_2 + G_3 + G_5) - V_A G_2 - E_2 G_3 = 0$$



KCL at Node 1:

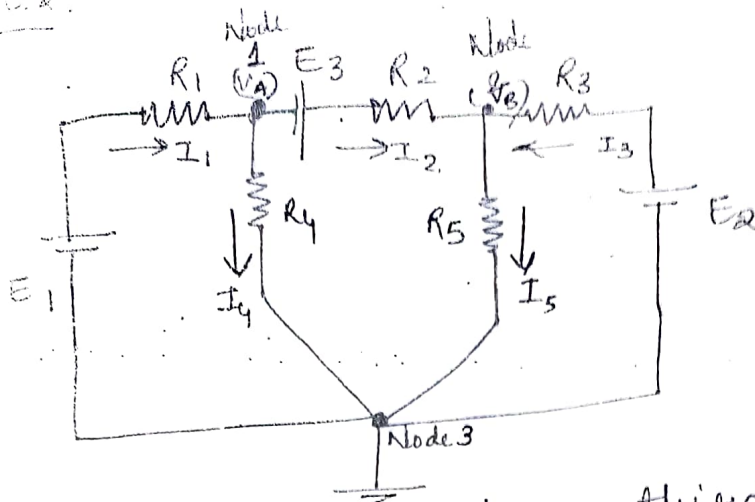
$$V_A \left( \frac{1}{R_1} + \frac{1}{R_5} + \frac{1}{R_2} + \frac{1}{R_8} \right) - \frac{E_1}{R_1} - \frac{V_B}{R_2} - \frac{V_C}{R_8} = 0$$

KCL at Node 2:

$$V_B \left( \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_6} \right) - \frac{V_A}{R_2} - \frac{V_C}{R_3} = 0$$

KCL at Node 3:

$$V_B \left( \frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_7} + \frac{1}{R_8} \right) - \frac{E_2}{R_4} - \frac{V_C}{R_3} - \frac{V_A}{R_8} = 0$$



Consider the case when a third battery of emf  $E_3$  is connected between nodes 1 and 2.

As we travel from node 1 to node 2, we go from the -ve terminal of  $E_3$  to its +ve terminal.

Hence,  $E_3$  must be taken as +ve. However, if we travel from node 2 to node 1, we go from the +ve to -ve terminal.

Apply KCL at node 1

$$I_1 = I_4 + I_2$$

$$E_1 - V_A = 0$$

$$\frac{E_1 - V_A}{R_1} = \frac{V_A}{R_4} + \frac{V_A + E_3 - V_B}{R_2}$$

$$\frac{E_1}{R_1} - \frac{V_A}{R_1} = \frac{V_A}{R_4} + \frac{V_A}{R_2} + \frac{E_3}{R_2} - \frac{V_B}{R_2} - \frac{E_1 + V_A}{R_1} = 0$$

$$V_A \left( \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4} \right) + \frac{E_3}{R_2} - \frac{E_1}{R_1} - \frac{V_B}{R_2} = 0$$

This is the same expression as given in first case except for the additional term involving  $E_3$ . This additional term is taken as  $+\frac{E_3}{R_2}$ .

(not as  $-\frac{E_3}{R_2}$ ) because when viewed from node 1, it represents rise in voltage.



Apply KCL at node 2

$$I_2 + I_3 = I_5$$

$$\frac{V_A + E_3 - V_B + E_2 - V_B}{R_2} = \frac{V_B}{R_5}$$

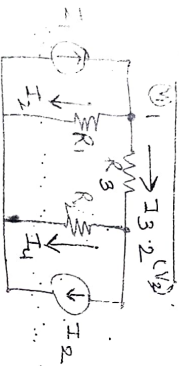
$$-\frac{V_A}{R_2} - \frac{E_3}{R_2} + \frac{V_B}{R_2} - \frac{E_2}{R_3} + \frac{V_B}{R_3} + \frac{V_B}{R_5} = 0$$

$$V_B \left( \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_5} \right) - \frac{V_A}{R_2} - \frac{E_2}{R_3} - \frac{E_3}{R_2} = 0$$

The additional term is  $-\frac{E_3}{R_2}$  (not  $+\frac{E_3}{R_2}$ ) because as viewed from node 3, it represents a fall in potential.

Nodal analysis with current sources

Consider a network



$$\frac{V_2}{R_2} + \frac{V_2}{R_3} - \frac{V_1}{R_3} = -I_2$$

$$V_2 \left( \frac{1}{R_2} + \frac{1}{R_3} \right) - \frac{V_1}{R_3} = -I_2 \quad (2)$$

Consider eqn (1)

(1) product of potential  $V_1$  and  $\left( \frac{1}{R_1} + \frac{1}{R_3} \right)$  i.e. sum of the reciprocals of branches resistance connected to this node.

(2) is the ratio of adjoining potential  $V_2$  and interconnecting resistance  $R_3$ .

(3) All the above equated to the current supplied by the current source connected to the node.

NOTE: Current is taken +ve, if flowing out of it, the node and -ve, if flowing in.

In terms of branch conductances

$$\text{Eqn (1)} \quad V_1(G_1 + G_3) - V_2 G_3 = I_1$$

$$V_1 G_1 + V_1 G_3 - V_2 G_3 = I_1$$

At node 1

$$I_1 = I_2 + I_3$$

$$I_1 = \frac{V_1}{R_1} + \frac{V_1 - V_2}{R_3}$$

$$I_1 = V_1 \left( \frac{1}{R_1} + \frac{1}{R_3} \right) - \frac{V_2}{R_3}$$

$$\frac{V_1}{R_1} + \frac{1}{R_3} - \frac{V_2}{R_3} = I_1 \quad (1)$$

At node (2)

$$I_3 = I_4 + I_2$$

$$\frac{V_1 - V_2}{R_3} = \frac{V_2}{R_2} + I_2$$

(or)

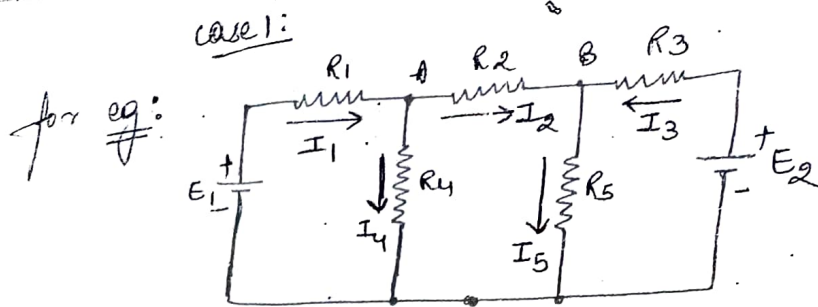
## NODE-VOLTAGE THEOREM ✓

For application of node voltage theorem one of the nodes is taken as reference or zero potential or datum node and the potential difference between each of the other nodes and the reference node is expressed in terms of an unknown voltage and at every node Kirchhoff's first (or current) law is applied assuming the positive directions of branch currents.)

This assumption does not change the statement of problem, since the branch currents are determined by the potential difference between respective nodes and not by absolute values of node potentials.

### Advantages:

- Like Maxwell's circulating current theorem, node-voltage theorem reduces the number of equations to be solved to determine the unknown quantities.
- If there are  $n$  no. of nodes, there shall be  $(n-1)$  number of nodal equations in terms of  $(n-1)$  no. of unknown variables of nodal voltages.
- By solving these eq<sup>n</sup> branch currents can be calculated.
- When  $(n-1)$  is less than the no. of independent meshes in the network, it is more advantageous.
- Moreover, it is particularly suited for networks having many parallel circuits with common ground conductive node such as in electronic circuits.
- These node equations often become simplified if all voltage sources are converted into current sources.
- Every junction where two or more elements meet together is considered as node.



At node A

$$I_1 = I_2 + I_4$$

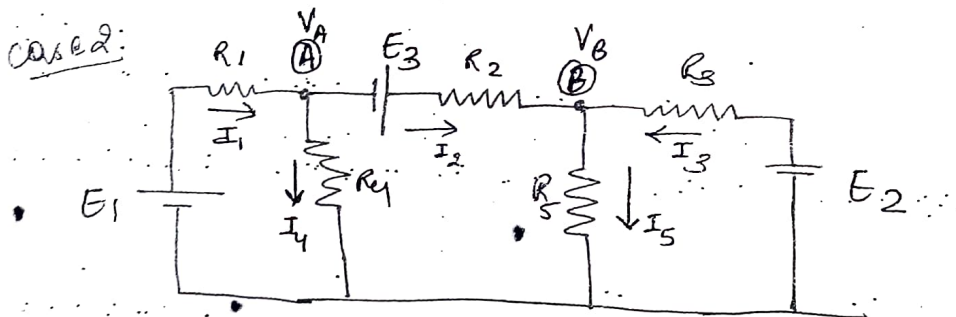
$$\frac{E_1 - V_A}{R_1} = \frac{V_A - V_B}{R_2} + \frac{V_A}{R_4}$$

At node B

$$I_2 + I_3 = I_5$$

$$\frac{V_A - V_B}{R_2} + \frac{E_2 - V_B}{R_3} = \frac{V_B}{R_5}$$

let at node A,  $V_A$  be the voltage  
node B,  $V_B$  be the voltage.



At node 1:

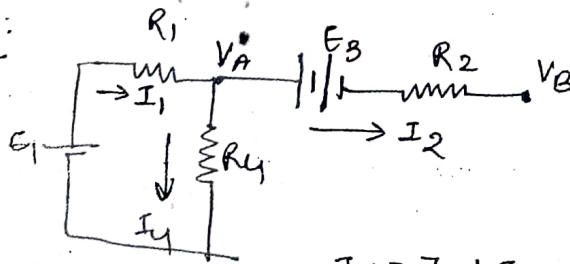
$$I_1 = I_2 + I_4$$

$$\frac{E_1 - V_A}{R_1} = \frac{V_A + E_3 - V_B}{R_2} + \frac{V_A}{R_4}$$

$$-V_A - E_3 + V_B = 0$$

$$V_A + E_3 = V_B$$

case 3:



$$I_1 = I_4 + I_2$$

$$\frac{E_1 - V_A}{R_1} = \frac{V_A}{R_4} + \frac{V_A - E_3 - V_B}{R_2}$$

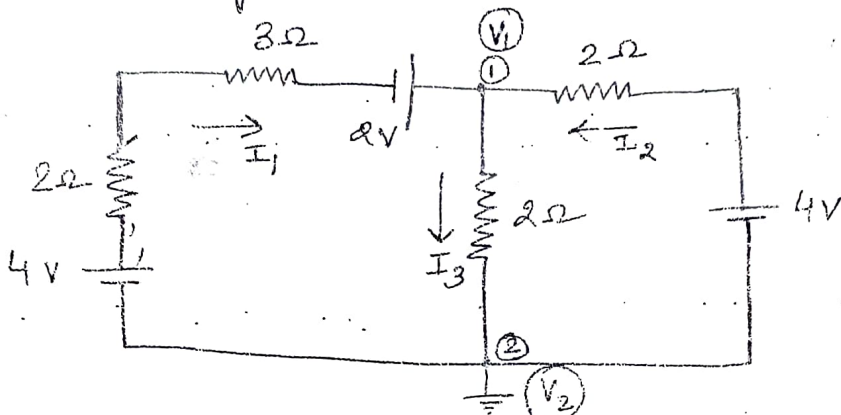
$$V_A + E_3 - V_B = 0$$

$$V_A = V_B - E_3$$



Q-1

Using Node Voltage method, find the current resistance for the network.



Soln

Let mark the directions of  $I_1, I_2$  &  $I_3$ .

$$I_1 + I_2 = I_3$$

$$\frac{4+2-V_1}{5} + \frac{4-V_1}{2} = \frac{V_1}{2}$$

$$\frac{6-V_1}{5} + \frac{4-V_1}{2} = \frac{V_1}{2}$$

$$\frac{2(6-V_1) + 5(4-V_1)}{10} = \frac{V_1}{2}$$

$$12 - 2V_1 + 20 - 5V_1 = 5V_1$$

$$32 - 7V_1 = 5V_1$$

$$32 = 5V_1 + 7V_1$$

$$32 = 12V_1$$

$$V_1 = \frac{32}{12} = \frac{8}{3}$$

$$= 6.67$$

$$V_1 = \frac{8}{3} \text{ V}$$

$$I_1 = \frac{4+2-V_1}{5}$$

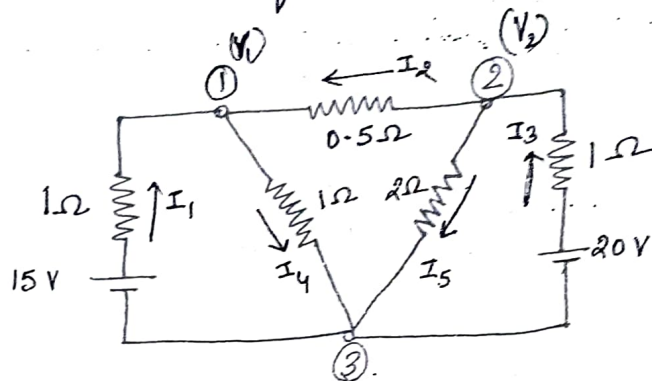
$$= \frac{6 - 8/3}{5} = \frac{18-8}{15}$$

$$= \frac{10}{15}$$

$$= 0.667 \text{ A}$$

Q2

frame and solve the node equations of the network.  
Hence, find the total power consumed by the passive elements of the network.



Soln

At node 1:

$$I_1 + I_2 = I_4$$

$$\frac{15 - V_1}{1} + \frac{V_2 - V_1}{0.5} = \frac{V_1}{1}$$

$$15 - V_1 + 2(V_2 - V_1) = V_1$$

$$15 - V_1 + 2V_2 - 2V_1 = V_1$$

$$15 + 2V_2 - 3V_1 - V_1 = 0$$

$$15 + 2V_2 - 4V_1 = 0 \quad \text{--- (3)}$$

$$4V_1 - 2V_2 = 15 \quad \text{--- (1)}$$

At node 2:

$$I_3 = I_2 + I_5$$

$$\frac{20 - V_2}{1} = \frac{V_2 - V_1}{0.5} + \frac{V_2}{2}$$

$$20 - V_2 = 2(V_2 - V_1) + \frac{V_2}{2}$$

$$20 - V_2 = \frac{4(V_2 - V_1) + V_2}{2}$$

$$2(20 - V_2) = 4V_2 - 4V_1 + V_2$$

$$40 - 2V_2 = 4V_2 - 4V_1 + V_2$$

$$40 = 4V_2 + 2V_2 + V_2 - 4V_1$$

$$40 = 7V_2 - 4V_1 \quad \text{--- (2)}$$

From eq<sup>n</sup> ①

$$4V_1 - 15 = 2V_2$$

$$V_2 = \frac{4V_1 - 15}{2}$$

Put 'V<sub>2</sub>' in eq<sup>n</sup>(2)

$$40 = 7\left(\frac{4V_1 - 15}{2}\right) - 4V_1$$

$$2 \times 40 = 7(4V_1 - 15) - 8V_1$$

$$80 = 28V_1 - 105 - 8V_1$$

$$80 + 105 = 20V_1$$

$$V_1 = \frac{185}{20} = 9.25$$

$$V_1 = \frac{37}{4} V$$

$$I_1 = \frac{15 - V_1}{1} = 15 - \frac{37}{4} = \frac{60 - 37}{4} = \frac{23}{4} A = 5.75 A$$

$$I_2 = 2(V_2 - V_1) = 2\left(11 - \frac{37}{4}\right) = 2\left(\frac{44 - 37}{4}\right) = \frac{7}{2} = 3.5 A$$

$$I_3 = 20 - V_2 = 20 - 11 = 9 A$$

$$I_4 = \frac{V_1}{1} = \frac{37}{4} = 9.25 A$$

$$I_5 = \frac{V_2}{2} = \frac{11}{2} = 5.5 A$$

Passive elements of n/w :

$$\begin{aligned} \text{Total power consumed by the passive element of n/w} \\ = (5.75)^2 \cdot 1 + (3.5)^2 \cdot (0.5) + (9)^2 \cdot 1 + (9.25)^2 \cdot 1 \\ + (5.5)^2 \cdot 2 \end{aligned}$$

$$= 33.0625 + 6.125 + 81 + 85.5625 + 60.5$$

$$= 266.25$$

$$P = 266.25 W$$

Put V<sub>1</sub> in eq<sup>n</sup>(2)

$$40 = 7V_2 - 4\left(\frac{37}{4}\right)$$

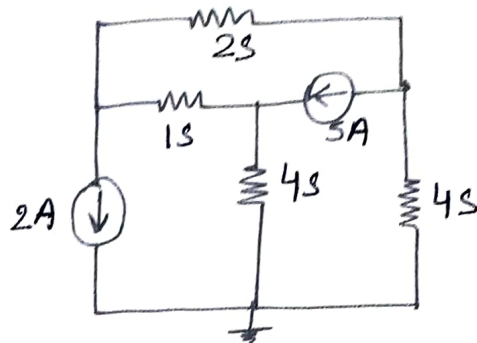
$$40 = 7V_2 - 37$$

$$40 + 37 = 7V_2$$

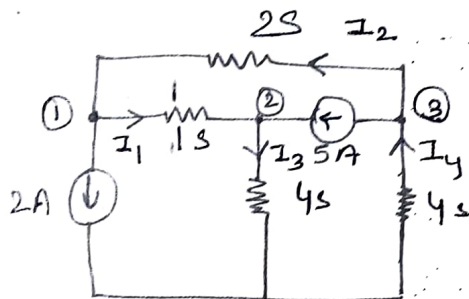
$$\frac{77}{7} = V_2$$

$$V_2 = 11 V$$

Q.3 Use nodal analysis method to find current in the various branches of the circuits. All branch conductances are given in siemens (mho).



Soln



KCL at node ①

$$\begin{aligned} I_2 &= 2 + I_1 \\ 2(V_3 - V_1) &= 2 + V_1 - V_2 \\ 2V_3 - 2V_1 &= 2 + V_1 - V_2 \\ 2V_3 - 2V_1 - V_1 + V_2 &= 2 \\ -3V_1 + V_2 + 2V_3 &= 2 \quad \text{--- (1)} \end{aligned}$$

KCL at node ②

$$\begin{aligned} I_1 + 5 &= I_3 \\ (V_1 - V_2) + 5 &= 4V_2 \\ -V_1 + V_2 + 4V_2 &= -5 \\ -V_1 + 5V_2 &= -5 \quad \text{--- (2)} \end{aligned}$$

KCL at node ③

$$\begin{aligned} I_4 &= 5 + I_2 \\ -V_3(4) &= 5 + (V_3 - V_1) 2 \\ -4V_3 &= 5 + 2V_3 - 2V_1 \\ -4V_3 - 2V_3 + 2V_1 &= 5 \\ 2V_1 - 6V_3 &= 5 \quad \text{--- (3)} \end{aligned}$$



Apply Cramer's rule:

$$\begin{bmatrix} -3 & 1 & 2 \\ -1 & 5 & 0 \\ 2 & 0 & -6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 5 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} -3 & 1 & 2 \\ -1 & 5 & 0 \\ 2 & 0 & -6 \end{bmatrix} = -3(-30) - 1(-6) + 2(0-10) \\ = 90 - 6 - 20 \\ = 64$$

$$\Delta = 64$$

$$\Delta_1 = \begin{bmatrix} 2 & 1 & 2 \\ 5 & 5 & 0 \\ 5 & 0 & -6 \end{bmatrix} = 2(-30) - 1(-30) + 2(0-25) \\ = -60 + 30 - 50 \\ = -80$$

$$\Delta_1 = -80$$

$$\Delta_2 = \begin{bmatrix} -3 & 2 & 2 \\ -1 & 5 & 0 \\ 2 & 5 & -6 \end{bmatrix} = -3(-30) - 2(-6) + 2(-5-10) \\ = 90 - 12 - 30 \\ = 48$$

$$\Delta_2 = 48$$

$$\Delta_3 = \begin{bmatrix} -3 & 1 & 2 \\ -1 & 5 & 5 \\ 2 & 0 & 5 \end{bmatrix} = -3(25) - 1(-5-10) + 2(0-10) \\ = -75 + 15 - 20 \\ = -80$$

$$\Delta_3 = -80$$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{-80}{64} = -1.25V$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{48}{64} = 0.75V$$

$$V_3 = \frac{\Delta_3}{\Delta} = \frac{-80}{64} = -1.25V$$

$$I_1 = V_1 - V_2$$

$$I_2 = 2(V_3 - V_1)$$

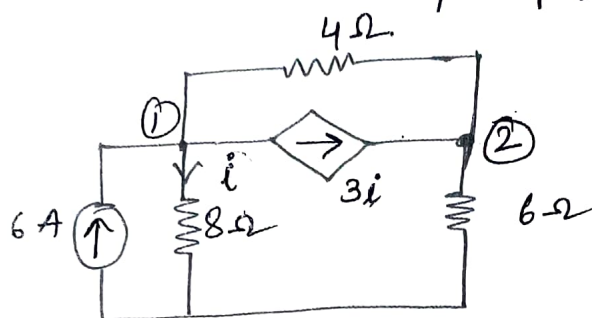
$$I_3 = 4V_2$$

$$I_4 = -V_3 \cdot 4$$

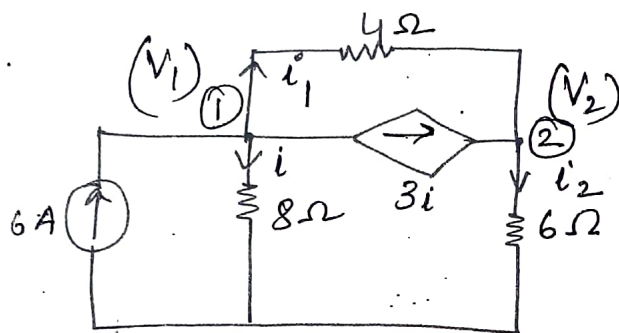
Q.4

Use nodal analysis to determine the value of current 'i' in the n/w of given fig.

Q-5.  
(BLT)



Soln



At node 1

$$6 = i + 3i + i_1$$

$$6 = 4i + \frac{V_1 - V_2}{4}$$

$$6 = 4\left(\frac{V_1}{8}\right) + \frac{V_1 - V_2}{4}$$

$$6 = \frac{V_1}{2} + \frac{V_1 - V_2}{4}$$

$$24 = 2V_1 + V_1 - V_2$$

$$24 = 3V_1 - V_2 \quad \text{--- (1)}$$

At node (2)

$$3i + i_1 = i_2$$

$$3\left(\frac{V_1}{8}\right) + \left(\frac{V_1 - V_2}{4}\right) = \frac{V_2}{6}$$

$$\frac{3V_1 + 2V_1 - 2V_2}{84} = \frac{V_2}{63}$$

$$3(5V_1 - 2V_2) = 4V_2$$

$$15V_1 - 6V_2 = 4V_2$$

$$15V_1 - 10V_2 = 0$$

$$3V_1 - 2V_2 = 0 \quad \text{--- (2)}$$

from eqn (1)

$$V_2 = 3V_1 - 24$$

Put  $V_2$  in eqn (2)

$$3V_1 - 2(3V_1 - 24) = 0$$

$$3V_1 - 6V_1 + 48 = 0$$

$$-3V_1 + 48 = 0$$

$$V_1 = \frac{48}{3}$$

$$\boxed{V_1 = 16V}$$

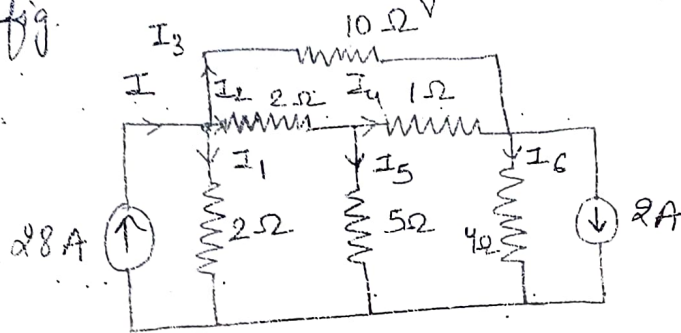
$$V_2 = 3(16) - 24$$

$$= 48 - 24$$

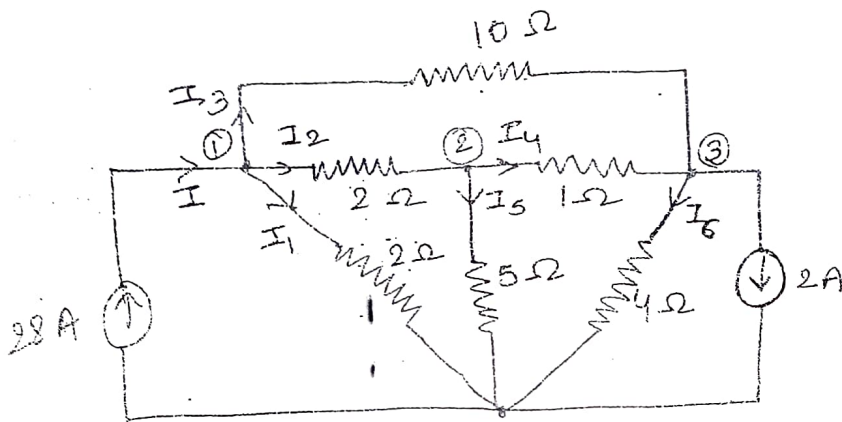
$$\boxed{V_2 = 24V}$$

$$i = \frac{V_1}{8} = \frac{16}{8} = 2A$$

Q-5. Use nodal analysis method to find current in the (BLT) various resistors of the circuit as shown in fig.



Soln.



Let voltage at node ① ② & ③ be  $V_1, V_2$  &  $V_3$ .

At node ①, Apply KCL

$$I = I_1 + I_2 + I_3$$

$$28 = \frac{V_1}{2} + \left( \frac{V_1 - V_2}{2} \right) + \left( \frac{V_1 - V_3}{10} \right)$$

$$28 = \frac{5V_1 + 5V_1 - 5V_2 + V_1 - V_3}{10}$$

$$280 = 11V_1 - 5V_2 - V_3 \quad \text{--- (1)}$$

At node ②,

$$I_2 = I_4 + I_5$$

$$\left( \frac{V_1 - V_2}{2} \right) = \left( \frac{V_2 - V_3}{1} \right) + \frac{V_2}{5}$$

$$5(V_1 - V_2) = 2(5V_2 - 5V_3 + V_2)$$

$$5V_1 - 5V_2 = 10V_2 - 10V_3 + 2V_2$$

$$5V_1 - 17V_2 + 10V_3 = 0 \quad \text{--- (2)}$$

At node (3)

$$I_4 + I_3 = I_6 + 2$$

$$\frac{V_2 - V_3}{1} + \frac{V_1 - V_3}{10} = \frac{V_3}{4} + 2$$

$$4(10V_2 - 10V_3 + V_1 - V_3) = 10(V_3 + 8)$$

$$40V_2 - 40V_3 + 4V_1 - 4V_3 = 10V_3 + 80$$

$$4V_1 + 40V_2 - 54V_3 = 80$$

$$2V_1 + 20V_2 - 27V_3 = 40 \quad \text{--- (3)}$$

Using Cramer's rule

$$\begin{bmatrix} 11 & -5 & -1 \\ 5 & -17 & 10 \\ 2 & 20 & -27 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 280 \\ 0 \\ 40 \end{bmatrix}$$

$$\begin{aligned} \Delta &= \begin{vmatrix} 11 & -5 & -1 \\ 5 & -17 & 10 \\ 2 & 20 & -27 \end{vmatrix} = 11(459 - 200) + 5(-135 - 20) - 1(100 + 134) \\ &= 11(259) + 5(-155) - (134) \\ &= 2849 - 775 - 134 \\ &= 2849 - 909 \\ \Delta &= 1940 \end{aligned}$$

$$\Delta_1 = \begin{bmatrix} 280 & -5 & -1 \\ 0 & -17 & 10 \\ 40 & 20 & -27 \end{bmatrix}$$

$$= 280(459 - 200) + 5(0 - 400) + 1(0 + 680)$$

$$= 280(259) - 2000 - 680$$

$$= 72520 - 2680$$

$$\Delta_1 = 69840$$

$$\Delta_2 = \begin{bmatrix} 11 & 280 & -1 \\ 5 & 0 & 10 \\ 2 & 40 & -27 \end{bmatrix} = 11(0 - 400) - 280(-125 - 20) - 1(200 - 0)$$

$$= 11(-400) - 280(-155) - 200$$

$$= -4400 + 43400 - 200$$

$$= -4400 + 43200$$

$$= 38800$$

$$\Delta_3 = \begin{bmatrix} 11 & -5 & 280 \\ 5 & -17 & 0 \\ 2 & 20 & 40 \end{bmatrix} = 11(-680) + 5(200) + 280(0 + 100) = -7480 + 1000 + 28000 = -7480 + 29000 = 31040$$

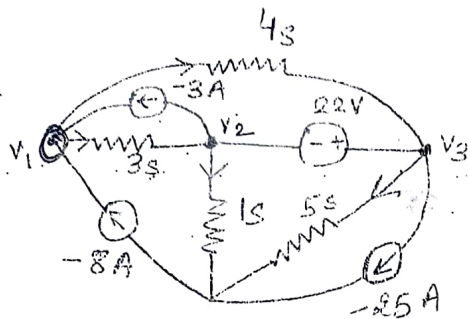
$$V_1 = \frac{\Delta_1}{\Delta} = \frac{69840}{1940} = 36 \text{ V}$$

$$V_2 = \frac{38800}{1940} = \frac{\Delta_2}{\Delta} = 20 \text{ V}$$

$$V_3 = \frac{\Delta_3}{\Delta} = \frac{31040}{1940} = 16 \text{ V}$$



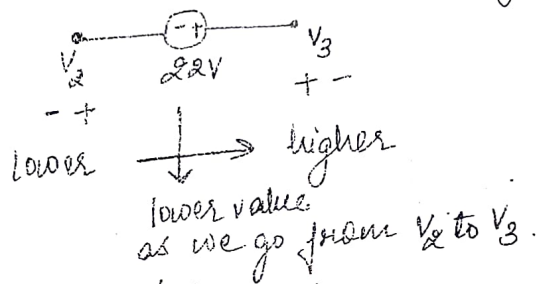
# SUPER NODE



Super node is one when one voltage source is present between two nodes only, no other elements are present.

So we cannot find out the value of  $V_2$  and  $V_3$ .

We can write down the total voltages for a line



So eqn for voltage will be.

$$-V_2 - 22 + V_3 = 0$$

$$V_2 + 22 - V_3 = 0 \quad \text{--- (1)}$$

At node ①  $V_1$ , apply KCL

$$(-3) + (-8) = (V_1 - V_2)3 + (V_1 - V_3)4$$

→ because values are given in siemens  
 $G = \frac{1}{R}$

$$(-3) + (-8) - (V_1 - V_2)3 - (V_1 - V_3)4 = 0$$

$$-11 - 3V_1 + 3V_2 - 4V_1 + 4V_3 = 0$$

$$-11 - 7V_1 + 3V_2 + 4V_3 = 0 \quad \text{--- (2)}$$

At node ②, apply KCL

$$3(V_1 - V_2) = -3 + V_2 \quad \text{--- (3)}$$

$$3(V_1 - V_2) + 3 - V_2 = 0 \quad \text{--- (3)}$$

At node ③, apply KCL

$$(V_1 - V_3)4 = -25 + 5V_3$$

$$(V_1 - V_3)4 + 25 - 5V_3 = 0 \quad \text{--- (4)}$$

Now if we remove the 22V voltage source then  $V_2$  &  $V_3$  can be considered as same node. So add eqn ③ & ④

$$3(V_1 - V_2) + 3 - V_2 + (V_1 - V_3)4 + 25 - 5V_3 = 0$$

$$3(v_1 - v_2) + 3 - v_2 + (v_1 - v_2)4 + 0.5 - 5v_2 = 0$$

$$3v_1 - 3v_2 + 3 - v_2 + 4v_1 - 4v_2 + 0.5 - 5v_2 = 0$$

$$7v_1 - 4v_2 - 9v_3 + 28 = 0 \quad \text{--- (5)}$$

Consider eq<sup>n</sup> (1), (2), (5)

$$v_2 - v_3 = -22 \quad \text{--- (1)}$$

$$-7v_1 + 3v_2 + 4v_3 = 11 \quad \text{--- (2)}$$

$$7v_1 - 4v_2 - 9v_3 = -28 \quad \text{--- (5)}$$

Using Cramer's rule

$$\begin{bmatrix} 0 & 1 & -1 \\ -7 & 3 & 4 \\ 7 & -4 & -9 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -22 \\ 11 \\ -28 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 0 & 1 & -1 \\ -7 & 3 & 4 \\ 7 & -4 & -9 \end{vmatrix} = 0 - 1(63 - 28) - 1(27 - 21) \\ = -(35) - (6) \\ = -41$$

$$\Delta_1 = \begin{vmatrix} -22 & 1 & -1 \\ 11 & 3 & 4 \\ -28 & -4 & -9 \end{vmatrix} = -22(-27 + 16) - 1(-99 + 112) - 1(-24 + 84) \\ = -22(-11) - 1(13) - (40) \\ = 242 - 13 - 40 = 242 - 53 = 189$$

$$\Delta_2 = \begin{vmatrix} 0 & -22 & -1 \\ -7 & 11 & 4 \\ 7 & -28 & -9 \end{vmatrix} = 0 - (-22)(63 - 28) - 1(196 - 77) \\ = 22(35) - (119) = 651$$

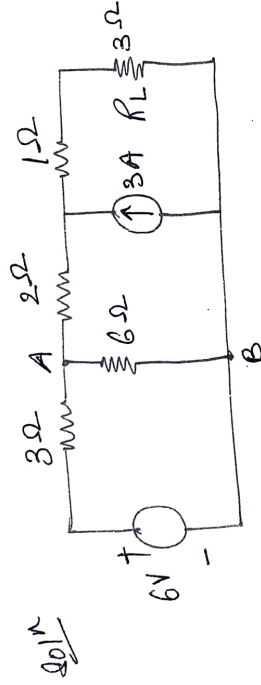
$$\Delta_3 = \begin{vmatrix} 0 & 1 & -22 \\ -7 & 3 & 11 \\ 7 & -4 & -28 \end{vmatrix} = 0 - 1(196 - 77) - 22(28 - 21) \\ = -(119) - 22(7) = -119 - 154 \\ = -273$$

$$v_1 = \frac{\Delta_1}{\Delta} = -4.5V \quad \left| \quad v_2 = \frac{\Delta_2}{\Delta} = -15.5V \quad \right| \quad v_3 = 6.5V$$

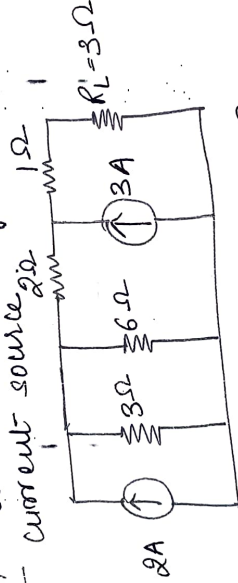
# NUMERICALS BASED ON SOURCE

## CONVERSION

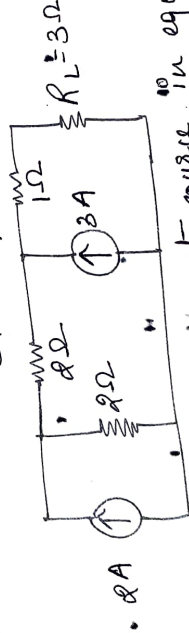
Q-1. Using source conversion technique to find the load current  $I$  in the circuit.



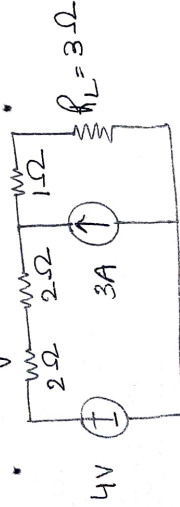
Step 1: Convert 6V voltage source in equivalent current source  $\frac{6}{3} = 2A$

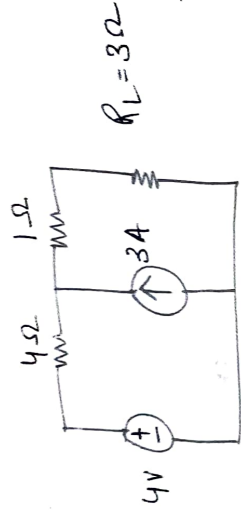


Step 2:  $3 \parallel 6 = \frac{3 \cdot 6}{3+6} = \frac{18}{9} = 2\Omega$

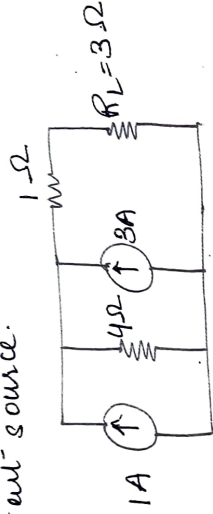


Step 3: Convert 2A current source in equivalent voltage source.

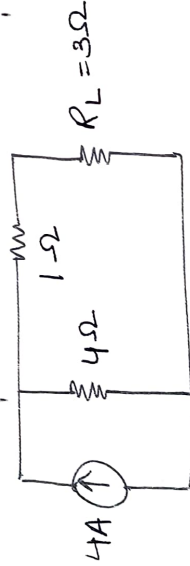




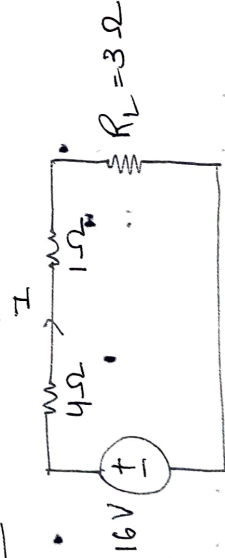
step 4: convert 4V voltage source in equivalent current source.



step 5:  $1A + 3A$



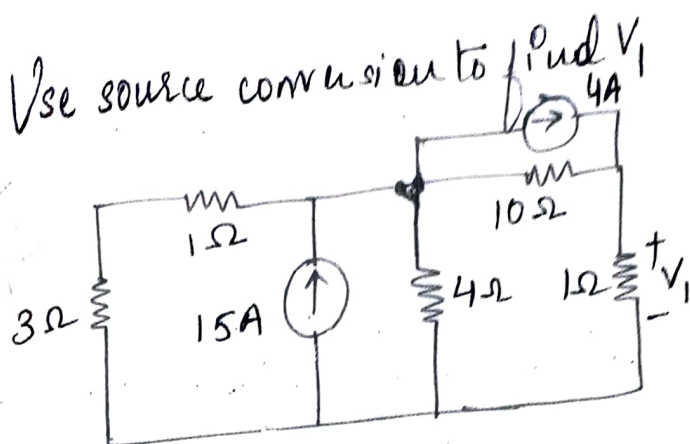
step 6: convert current source



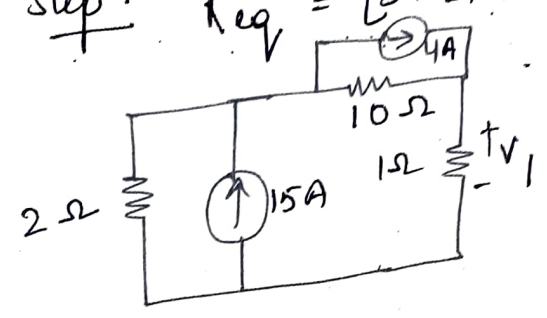
$$I = \frac{16}{8} = 2A$$

$$\boxed{I = 2A}$$

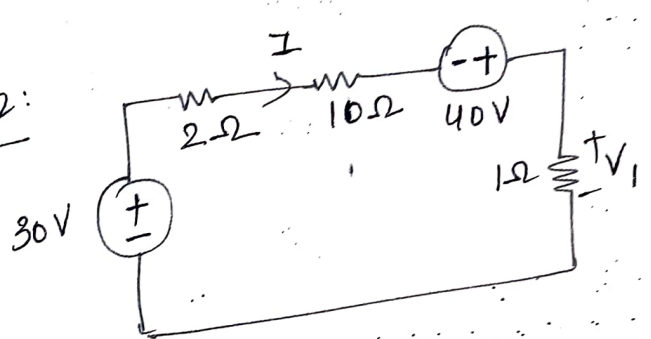
Q2. Use source conversion to find  $V_1$



Sol<sup>n</sup> Step 1:  $R_{eq} = [3+1] \parallel 4 = 4 \parallel 4 = 2 \Omega$



Step 2:



$$-30 + 2I - 40 + I = 0$$

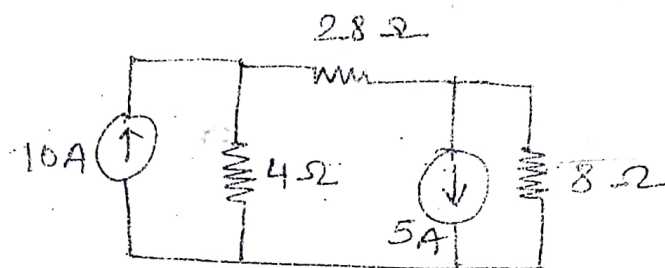
$$13I = 70$$

$$I = \frac{70}{13}$$

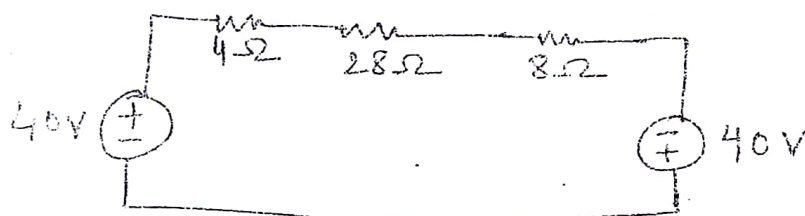
$$V_1 = \frac{70}{13} \times 1 = \frac{70}{13} = 5.38V$$



Q3. Using source conversion, determine the current in  $28\Omega$  resistor in the circuit



Sol<sup>n</sup>



KVL

$$-40 + I(4 + 28 + 8) - 40 = 0$$

$$-80 + I[40] = 0$$

$$I = \frac{80}{40} = 2$$

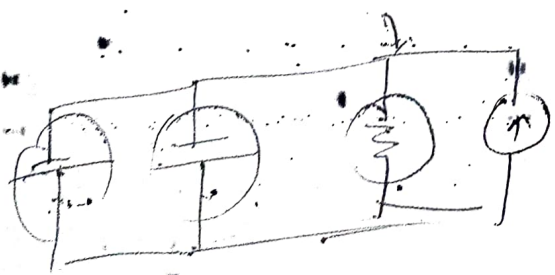
$$I = 2A$$

# SUPERPOSITION THEOREM

Superposition theorem can be stated as below:  
 In a linear resistive network containing two or more voltage sources, the current through any element (resistance or source) may be determined by adding together algebraically the currents produced by each source acting alone, when all other voltage sources are replaced by their internal resistances. If a voltage source has no internal resistance, the terminals to which it was connected are joined together. If there are current sources present they are removed and the network terminals to which they were connected are left open."

2. In a network of linear resistances containing more than one generator (or source of emf), the current which flows at any point is the sum of all the currents which would flow at that point if each generator were considered separately and all the other generators replaced for the time being by resistances equal to their internal resistances.

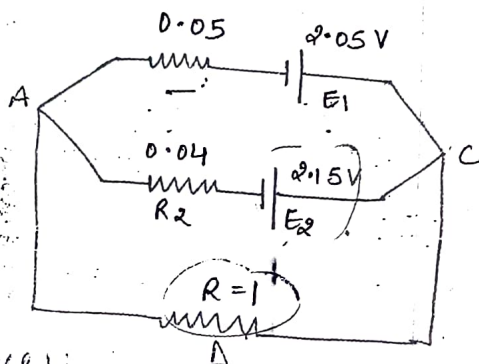
$$r_1 + r_2 = r_{eq}$$



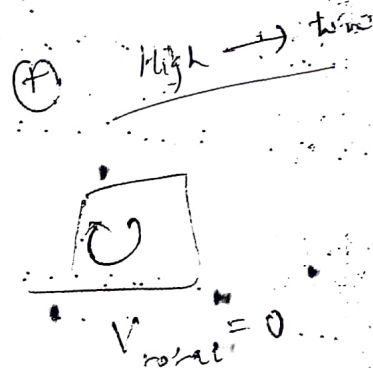
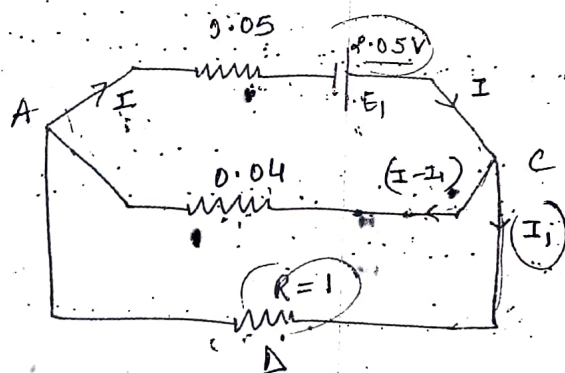
# NUMERICALS BASED ON SUPERPOSITION THEOREM

Q.1. By using superposition theorem, find the current in resistance  $R$  shown in fig.

$R_1 = 0.005 \Omega$   $R_2 = 0.004 \Omega$ ,  $R = 1 \Omega$ ,  $E_1 = 2.05V$ ,  $E_2 = 2.15V$   
Internal resistances of cells are negligible.



Soln  
Case 1:  
consider  $E_1$  source at one time and short circuit the other sources.  
Ckt. will be



Apply KVL in ACA,

$$0.05I - 2.05 + 0.04(I - I_1) = 0$$

$$0.05I - 2.05 + 0.04I - 0.04I_1 = 0$$

$$0.09I - 0.04I_1 = 2.05 \quad \text{--- (1)}$$

Apply KVL in ACBA,

$$0.04(I - I_1) - I_1 = 0$$

$$0.04I - 0.04I_1 - I_1 = 0$$

$$0.04I - 1.04I_1 = 0 \quad \text{--- (2)}$$